Evaluation of Magnetic Rotor Position Sensors for PMSM Control in Electric Vehicle Applications

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***Abstract*—Permanent Magnet Synchronous Motors (PMSMs) have become a key technology in modern electric vehicles due to their high efficiency, power density, and torque performance. Accurate rotor position information is essential for Field-Oriented Control (FOC), as position estimation errors can lead to torque ripple, reduced efficiency, increased acoustic noise, and degraded drive performance. Consequently, the selection of an appropriate rotor position sensing technology remains a critical design consideration in electric vehicle traction systems.**

This paper presents a detailed review and comparison of magnetic rotor position sensors used in PMSM drives, with a particular focus on Hall-effect, Anisotropic Magnetoresistive (AMR), and Giant Magnetoresistive (GMR) sensors. The operating principles, sensing mechanisms, performance characteristics, and practical implementation challenges of each technology are examined. Furthermore, their effectiveness in rotor position estimation, torque ripple mitigation, environmental robustness, temperature tolerance, and suitability for automotive applications is evaluated.

The analysis highlights the trade-offs between cost, accuracy, sensitivity, and system complexity associated with each sensing approach. While Hall-effect sensors remain the preferred solution for cost-sensitive applications, AMR and GMR sensors offer significantly improved position estimation accuracy and control performance, making them attractive for next-generation electric vehicle platforms. In addition, emerging developments in sensorless control, adaptive observers, machine learning-based estimation, and sensor fusion techniques are discussed as promising directions for future research. The findings provide valuable insights for the design and selection of rotor position sensing systems in high-performance and reliable electric vehicle propulsion applications.

***Keywords*—Permanent Magnet Synchronous Motor (PMSM), Electric Vehicles, Rotor Position Sensing, Hall-Effect Sensor, Anisotropic Magnetoresistance (AMR), Giant Magnetoresistance (GMR), Field-Oriented Control, Torque Ripple, Sensorless Control**

I. INTRODUCTION

As the world is moving forward towards the ”Net Zero Carbon Standard” Goal set by the United Nations, the reliance of the Automotive Sector on Fossil Fuels and ICE(Internal Combustion Engines) creates an obstacle in achieving this goal. EV(Electrical Vehicles) in contrast to ICE uses ”Electrical Motor Unit” like PMSM(Permanent Magnet Synchronous Motors), which has a lower negative environmental impact in comparison to the use of Fossil Fuels. In addition to environmental benefits, PMSM can also generate more torque in comparison to ICE, and can provide uplifted performance [1].

PMSMs are conventionally controlled using Rotor Field-Oriented Control (FOC). This control strategy requires accurate knowledge of the rotor position in order to properly align the stator current vector with the rotor magnetic field. The three-phase stator currents are transformed into direct-axis (d-axis) and quadrature-axis (q-axis) components, where the q-axis current primarily produces torque while the d-axis current influences the motor flux [2].

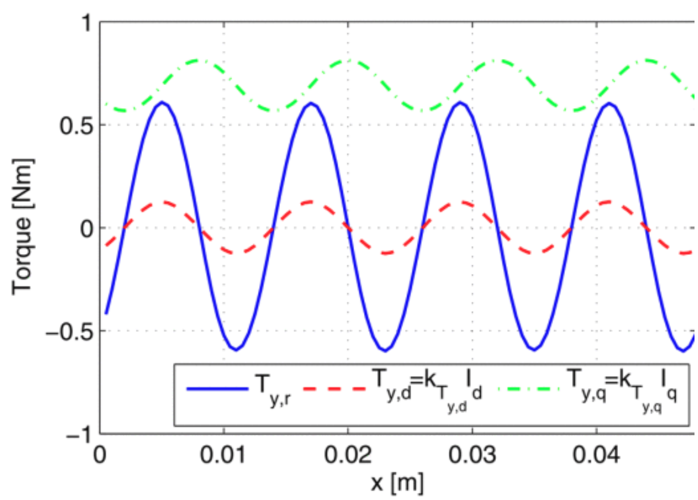

Fig. 1. Dependence of Torque on Rotor Position [2]

The electromagnetic torque produced by a PMSM can be decomposed into a reluctance component $(T_{y,r})$, a d-axis component $(T_{y,d})$, and a q-axis component $(T_{y,q})$. These torque components are highly dependent on the position of the moving translator x which is directly related to the mechanical position of the rotor, as can be observed in Fig. 1.

As Force and Torque control in a PMSM is dependent on the continuous position information of the motor rotor, any imprecision in its value can result in torque ripple, noise and vibration in performance [3]. Torque Ripple is a problem, where PMSM motor, due to inconsistency of the position of the rotor, uneven torque is generated at start or at rapid acceleration [4]. Such problems can be solved if exact position of the rotor can be known or calculated. In such cases use of Rotor Position sensors is necessary.

For a sensor to function adequately in the Automotive Sector, it must be unaffected by harsh conditions which can be internal like oil leakage, electromagnetic interference, etc. as well as external like extreme temperature, heavy rain, etc. [5]. Sensors must also be cheap and easily available, which can maintain the vehicles cost of manufacturing and be easily available in case of a fault [5].

The automotive industry has various PMSM drive rotor position sensing technologies like magnetic rotational angle (MR) sensors, optical rotational sensors, variable reluctance sensors, etc. MR sensors like Hall-Effect sensor, Anisotropic Magnetoresistive (AMR) and Giant Magnetoresistive (GMR) have become particularly attractive due to their compatibility with electric vehicles. Despite significant progress in rotor position estimation existing methods still suffer from a trade-off between cost, accuracy, and robustness. In particular, accurate rotor position estimation under sensor misalignment, low-speed operation, and varying load conditions remains an open challenge.

In this paper, we analyze the working principle, and effectiveness in mitigating torque ripple as well as the suitability for electric vehicle applications of various magnetic position sensors for PMSM motors and provide a comparative analysis between them, along with discussions about the future trends of sensorless position estimation.

II. Analysis of Magnetic Rotor Position Sensors

Magnetic field sensing has evolved from early compass-based direction finding to highly integrated solid-state devices capable of precise vector field measurement. Modern magnetometers are broadly used to measure magnetic field's magnitude and direction, and are now key components in applications ranging from navigation and geophysical exploration to industrial automation and electric drive systems. With advancements in semiconductor and thin-film technologies, magnetic sensors have become compact, cost-effective, and suitable for high-volume automotive integration, enabling their widespread adoption in motor control applications [6].

In electric vehicle traction systems, particularly PMSM drives, magnetic rotor position sensing is a critical function, as it provides the feedback required for accurate current commutation and torque production. The performance of these sensors directly influences control quality, especially in terms of torque ripple, efficiency, and low-speed operation stability. As a result, several sensing technologies have been developed and optimized to estimate rotor position with varying trade-offs in accuracy, cost, robustness, and environmental tolerance.

Based on their sensing principles and performance characteristics, magnetic rotor position sensors used in automotive applications can be broadly classified into Hall-effect sensors and magnetoresistive sensors, including Anisotropic Magnetoresistive (AMR) and Giant Magnetoresistive (GMR) types. The following sections analyze these technologies in detail, focusing on their operating principles, advantages, limitations, and suitability for torque ripple reduction and automotive deployment.

A. Hall-Effect Sensors

Hall Effect Sensors are the cheapest and most readily available sensors for Rotor Postion Sensing in the Industry. Hall Effect Sensors works on the principle of Hall Effect, which is in a way caused by Lorentz Force as shown in Fig. 2. When Magnetic Field applied perpendicularly to the electric current, the carrier carrying the current experience the Lorentz Force, which eventually result in generation of Hall Effect [7].

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}) \tag{1}$$

where:

$\mathbf{F}$: Lorentz force acting on the charge carrier (N)
q : Electric charge of the carrier (C)
$\mathbf{v}$: Velocity of the charge carrier (m/s)
$\mathbf{B}$: Magnetic flux density (T)
$\times$: Vector cross-product operator

The PMSM motors consists of Rotor Magnets. As Rotor Magnet's North and South Pole moves according to the rotation required in the PMSM Motor [8]. The Hall Effect Sensors are applied perpendicularly to the PMSM motor Rotor. The Rotor continuously rotates, and depending on the magnetic pole passed, the hall voltage is generated. [9] The Hall voltage is denoted by

$$V_y = k_H B_z J_x d_y \tag{2}$$

where:

V_y : Hall voltage (V)
k_H : Hall coefficient (m^3/C)
B_z : Magnetic flux density perpendicular to the current flow (T)
J_x : Current density flowing through the Hall element (A/m^2)
d_y : Distance between the Hall voltage sensing electrodes (m)

As the electrons are accumulated at the edge of the charge carrier by the magnetic force, it generates the hall voltage [9]. Hall Effect on the sensor is generated by Rotor Magnets in PMSM. Depending on the magnetic flux, the voltages are determined in the postion of the Rotor [10].

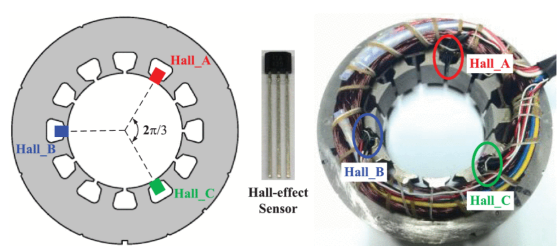

Fig. 2. Working of Hall Effect Sensor [11]

The main advantage of Hall sensors is derived from its design. Between the hall sensor and rotor, there is an air gap. Hence, Hall sensors in contactless mode can help is providing accurate measure of the rotor position [12]. In comparison to optical encoders, Magnetic sensors like hall effect perform better under improper lighting conditions [5]. Hall Sensors can also perform well in high magnetic field of strong permanent magnet in PMSM without any saturation. Hall sensors are manufactured using standard fabrication for semiconductors, which results in low cost [12]. Hall effect sensors can also provide enhanced performance if Ferromagnetic materials like Iron, Nickel, Cobalt, etc are used for fabrication, which lead to the generation of Anomalous Hall Effect [13].

Apart from various advantages, Hall effect sensors face the hurdle of limited sensitivity in comparison to other magnetic rotor position sensors. This can result in poor performance in case of weak magnetic fields [12]. Limited temperature resistance of Hall Effect Sensors is not suitable for harsh conditions. Lower operating frequency and slower response time, in ambient temperature adds an additional challenge in time critical Automotive Domain [14].

B. AMR

Anisotropic Magnetoresistive (AMR) sensors operate based on the anisotropic magneto-resistance effect, where the electrical resistance of a ferromagnetic material changes according to the orientation of an external magnetic field relative to its magnetized thin film, which is typically composed of nickel-iron (NiFe) permalloy, as illustrated in Fig. 3 [6]. In most implementations, either a quadruple layer sensor is used or four single layer AMR sensing elements are connected in a Wheatstone bridge configuration. [15] Sine and Cosine functions can be used on the measured voltages to determine the rotor angle position. As the magnetic field generated by a rotating permanent magnet changes direction, the resistance of the sensing elements varies, producing an output voltage proportional to the field orientation. Unlike Hall-effect sensors, AMR sensors primarily respond to the direction of the magnetic field rather than its strength [6]. The use of integrated thin-film technology also allows signal-conditioning circuitry to be fabricated on the same chip, resulting in compact sensor solutions [6].

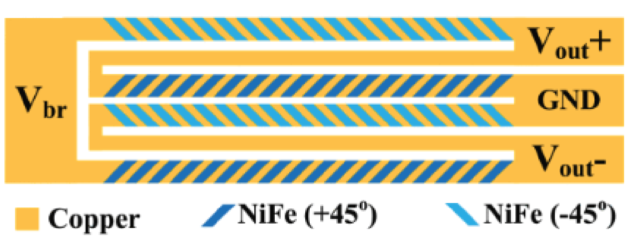

Fig. 3. Sensing element of AMR Sensor [6]

AMR sensors offer several advantages for rotor position sensing in PMSM-based electric vehicle drives. They are more sensitive than Hall-effect sensors and exhibit superior offset stability since they are not affected by the piezoelectric effect [16]. Furthermore, AMR sensors can operate at zero speed, provide rotation-direction detection, and offer excellent measurement repeatability [17]. Their ability to function with air gaps of up to approximately 3 mm and withstand operating temperatures of up to 200°C makes them particularly suitable for harsh automotive environments [17].

Despite these advantages, AMR sensors also present certain drawbacks. Compared to Hall-effect sensors, they generally have a higher cost and require active bias currents for operation, leading to increased power consumption [17]. In addition, their signal-conditioning circuitry can increase implementation complexity, which may be a consideration in cost-sensitive applications.

C. GMR

Giant Magnetoresistive (GMR) sensors operate based on the giant magneto-resistance effect, where the electrical resistance of a multilayer structure changes significantly in response to an external magnetic field. Usually GMR sensors are composed of ferromagnetic and non-magnetic conductive layers. These layers are alternating and are arranged in a way that one magnetic layer maintains a fixed direction of magnetization, called as the pinned layer, and another magnetic layer can freely change its magnetization as per the direction of external magnetic field, called as the free layer [6]. The sensor performs with a low resistance when there is parallelism between the magnetization direction of these layer and high resistance when there is anti-parallelism [16]. Just like AMR Sensors, the sensing elements of AMR sensor are usually arranged in a Wheatstone bridge configuration, and sine and cosine functions on these voltages eventually help determine the rotor angle position, which improves the sensitivity and also provides compensation against the temperature variations [16].

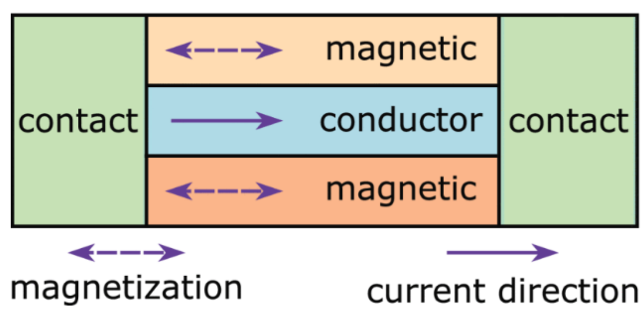

Fig. 4. Sensing element of GMR Sensor [6]

In PMSM rotor position sensing applications, a permanent magnet attached to the rotor produces a magnetic field whose orientation changes with rotor movement. The changing field alters the magnetization direction of the free layer, resulting in a measurable resistance variation that can be used to estimate rotor position accurately [16]. Due to their high sensitivity, GMR sensors are particularly suitable for precise angular measurements and rotor position detection in electric vehicle motor drives [17].

One of the major advantages of GMR sensors is their superior sensitivity compared to both Hall-effect and AMR sensors. At room temperature, GMR sensors typically provide three to six times higher sensitivity than AMR sensors, enabling more accurate detection of small magnetic field variations [17]. They also support operation at zero speed, provide excellent rotational measurement repeatability, allow larger air-gap operation of up to approximately 3.5 mm, and can be integrated with signal-conditioning circuitry on the same chip [17]. Furthermore, their compact size and high spatial resolution make them attractive for modern automotive applications [16].

Despite these advantages, GMR sensors also have several limitations. They require precise biasing and signal-conditioning electronics to achieve optimal performance, increasing implementation complexity [17]. Additionally, strong magnetic fields and elevated temperatures may affect the magnetization of the pinned layer, potentially degrading sensor performance [16]. GMR sensors also exhibit relatively higher noise levels compared to some alternative magnetic sensing technologies [16]. Therefore, while GMR sensors offer excellent accuracy and sensitivity for PMSM rotor position estimation, their cost, complexity, and environmental robustness must be carefully considered when selecting sensors for electric vehicle applications.

D. Comparison

From a torque ripple perspective, Hall-effect sensors provide the least effective mitigation because of their relatively low angular resolution and discrete position measurements. AMR sensors offer improved torque ripple reduction through more accurate and stable rotor position estimation. GMR sensors provide the best performance for torque ripple minimization due to their superior sensitivity and measurement accuracy, allowing the controller to maintain a more precise alignment between the stator and rotor magnetic fields [16] [17]. AMR sensors are limited to a 180° native sensing range, they introduce a startup position ambiguity that can cause motor jerking or reverse-rotation errors if not corrected by the controller. Conversely, while Hall effect and GMR sensors provide unambiguous 360° absolute tracking at startup, they lack the superior, sub-degree angular precision that AMR sensors deliver during continuous high-speed operation.

Considering automotive feasibility, Hall-effect sensors remain the most widely adopted solution because of their low cost, proven reliability, and ease of integration. For electric vehicle traction systems that need higher accuracy for control, AMR sensors are a balanced compromise between performance and robustness. GMR sensors deliver the highest sensing performance and greatest potential for torque ripple reduction; however, their higher cost and increased implementation complexity may limit their use to high-performance or premium electric vehicle applications. A detailed comparison of the three magneto-resistive sensors for rotor position sensing is presented in Table 1.

Overall, Hall-effect sensors are the most economical and automotive-proven solution, AMR sensors provide an excellent balance between accuracy and robustness, and GMR sensors offer the highest position estimation accuracy and torque ripple reduction capability at the expense of increased complexity and cost.

TABLE I
COMPARISON OF MAGNETIC SENSORS FOR ROTOR POSITION SENSING

Parameter	**Hall**	**AMR**	**GMR**
Sensitivity	Low	High	Very High
Angular Precision	60°	$\leq 0.1°$	$0.1° - 0.5°$
Native Angle Range	360°	180°	360°
Operation at Zero Speed	Limited	Yes	Yes
Cost	Low	Medium	High
Temperature Capability	Up to 125°C	Up to 200°C	Up to 240°C
Air Gap Capability	0.5mm-2mm	∼3 mm	∼3.5 mm
Environmental Drawbacks	Poor lighting	EMI	Temp.
Power Consumption	Low	High	Moderate
Bandwidth	30kHz	5mHz	1mHz
Fabrication	Standard	Add. Circuitry	Complex
Response Time	Moderate	Fast	Very Fast

III. FUTURE WORKS

Future research on rotor position sensing for electric vehicle traction systems is expected to focus on improving accuracy, reliability, and fault tolerance while reducing system cost and complexity. One promising direction is the enhancement of low-resolution magnetic sensing technologies through advanced estimation algorithms. Recent work has demonstrated that compensation techniques can significantly improve the position estimation capability of low-resolution Hall sensors, suggesting that future developments may enable low-cost sensing solutions to achieve performance levels comparable to more expensive encoder and resolver systems [18].

Another important area of research is the development of more robust sensorless control strategies. Although observer-based estimation methods have shown good performance under normal operating conditions, their accuracy can still be affected by parameter variations, phase delays, and model uncertainties. Future work should therefore focus on adaptive observers and real-time compensation techniques that can maintain accurate rotor position estimation across a wide speed range and under varying operating conditions [19] [20] .

Accurate rotor position estimation at standstill and low speeds remains a significant challenge for sensorless drives. Further investigation of high-frequency signal injection techniques may improve startup performance and low-speed controllability while reducing torque ripple, acoustic noise, and computational complexity. The continued refinement of these methods could contribute to more reliable sensorless operation throughout the entire speed range of electric vehicle motors [21].

Artificial intelligence is also expected to play an increasingly important role in future position estimation systems.

Machine learning models have demonstrated the ability to estimate rotor position directly from motor electrical signals, reducing dependence on precise motor models and parameter identification procedures. Future research may focus on developing lightweight neural-network architectures capable of real-time implementation on embedded motor controllers, as well as improving their robustness under changing operating conditions [22].

A particularly promising research direction is the integration of conventional estimation techniques with artificial intelligence. Hybrid approaches combining observer-based methods with machine learning models may exploit the advantages of both data-driven and physics-based estimation, leading to improved accuracy, robustness, and adaptability in practical applications [20] [22].

Finally, future electric vehicle drive systems are expected to increasingly employ sensor fusion and fault-tolerant estimation frameworks. By combining information from multiple sensing and estimation sources, it may be possible to maintain reliable rotor position information even in the presence of sensor degradation or failure. Such self-adaptive and fault-resilient architectures could improve the safety, reliability, and operational availability of next-generation electric vehicle propulsion systems [19].

IV. Conclusion

In this paper, a review and comparison of magnetic rotor position sensors presented, which are used in PMSM used in EVs. In order to tackle Torque Ripple, accurate rotor position is required for FOC. With this, torque production, efficiency of the PMSM and challenges like Torque Ripple can be managed. Selection of the type of Rotor Position System is crucial and is based on various factors like Cost, Temperature Drift, Sensitivity to Magnetic Flux, etc.

Among all magnetic Rotor Position Sensors reviewed, Hall Effect Sensors are still widely used due to their low production cost, less complexity in fabrication, and reliability in automotive environments. However, even though widely used, Hall Effect sensors faces the challenge of low sensitivity and limited angular resolution, in some scenarios may malign the position readings. In comparison to Hall Effect Sensors, AMR Sensors, working on the principle of Magnetoresistance, offers better sensitivity and repeatability, high temperature tolerance, with a drawback of increased fabrication complexity. GMR sensors in contrast, provide highest sensitivity and position estimation accuracy among the rest of the Magnetic Rotor Position Sensors which are reviwed earlier, although these benefits are accompanied by increased cost and more complex signal-conditioning requirements.

The comparative analysis demonstrates that no single sensor technology can be considered universally optimal for all electric vehicle applications. Instead, the choice of sensor depends on the specific requirements of the drive system, including accuracy, robustness, environmental tolerance, implementation complexity, and cost constraints. While Hall-effect sensors continue to dominate cost-sensitive applications, the growing demand for higher efficiency and smoother torque production is expected to increase the adoption of AMR and GMR technologies in future electric vehicle platforms.

The recent developments in the field of sensorless control techniques and position detection based on machine leaning algorithms with sensor fusion technology indicate a shift towards intelligent motor position estimation systems. These approaches aim to improve estimation accuracy, enhance fault tolerance, and reduce dependence on dedicated hardware sensors. In the future, motor position sensors for electric vehicle will likely combine advanced magnetic sensing technologies with data driven and intelligent systems, to achieve greater efficiency, reliability, and operational robustness while minimizing overall system cost.

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